

YU WANG

PhD Candidate · *Institute for Marine and Antarctic Studies, Hobart, Australia*

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RESEARCH INTERESTS

Antarctic ice dynamics	Ice sheet / ice shelf modelling	Subglacial hydrology
Glacial climate intervention	Climate and sea-level change	Ice–ocean interactions

EDUCATION

Doctor of Philosophy in Marine and Antarctic Studies 2022 - present
University of Tasmania, Australia

Bachelor of Marine and Antarctic Science with Honours (First Class) 2019 - 2021
University of Tasmania, Australia

Bachelor of Marine Science 2017 - 2021
Ocean University of China, China

PUBLICATIONS

Submitted / In-Preparation

- Beckmann, J., Reese, R., ... **Wang, Y.**, ... et al. [Disentangling uncertainty in ISMIP6 Antarctic sub-shelf melting and 2300 sea level rise projections](#). (Under review in *The Cryosphere*).
- King, M. A., Bertler, N. A. N., ... **Wang, Y.**, ... et al. Dynamic instability of the Wilkes Subglacial Basin Ice Sheet, East Antarctica. (Submitted to *Nature Reviews Earth & Environment*).
- Jager, E., Gladstone, R., Zwinger, T., **Wang, Y.**, Zhao, C., & Uotila, P. Insights from an ensemble inverse method to quantify basal friction uncertainties for ice sheet modelling - Part 2: Case of the Antarctic ice sheet. (Submitted to *Journal of Geophysical Research - Earth Surface*).
- **Wang, Y.**, Zhao, C., Gladstone, R., Zwinger, T., Galton-Fenzi, B. K., Christoffersen, P., & Jager, E. Future irreversible ice loss in East Antarctica accelerated by subglacial hydrology. (In prep.).
- **Wang, Y.**, Gladstone, R., Zhao, C., Wolovick, M., Moore, J., Christoffersen, P., Galton-Fenzi, B. K., & Jager, E. Self-regulating subglacial hydrology limits bed-drying stabilization in East Antarctica. (In prep.).
- Zhao, C., King, M., Hopuare, M., Maamaatuaiahutapu, K., & **Wang, Y.** Sea-level rise fingerprints from the Antarctic Ice Sheet on Pacific Islands. (In prep.).

Published

- Zhao, C., Gladstone, R., Zwinger, T., Gillet-Chaulet, F., **Wang, Y.**, Caillet, J., Mathiot, P., Saraste, L., Jager, E., Galton-Fenzi, B. K., Christoffersen, P., & King, M. A. (2025). [Subglacial water amplifies Antarctic contributions to sea-level rise](#). *Nature Communications*, 16, 3187.
- **Wang, Y.**, Zhao, C., Gladstone, R., Zwinger, T., Galton-Fenzi, B. K., & Christoffersen, P. (2024). [Sensitivity of the future evolution of the Wilkes Subglacial Basin ice sheet to grounding-line melt parameterizations](#). *The Cryosphere*, 18, 5117–5137.
- Seroussi, H., ... **Wang, Y.**, ... & Zwinger, T. (2024). [Evolution of the Antarctic Ice Sheet over the](#)

next three centuries from an ISMIP6 model ensemble. *Earth's Future*, 12(9), e2024EF004561.

- Wang, Y., Zhao, C., Gladstone, R., Galton-Fenzi, B., & Warner, R. (2022). [Thermal structure of the Amery Ice Shelf from borehole observations and simulations](#). *The Cryosphere*, 16, 1221–1245.

AWARDS & FUNDING

Travel Grant, ICEMAP Project, University of Lapland	2024
Travel Grant, Scientific Committee on Antarctic Research (SCAR)	2024
Travel Grant, Australian National University (ANU)	2023
Best Poster Award, ACCESS Community Workshop	2023
Australian Antarctic Program Partnership Top-Up Scholarship	2022 - present
Tasmanian Graduate Research Scholarship	2022 - present
Executive Dean's Honour Roll, University of Tasmania	2021
Tasmanian International Scholarship	2019 - 2021
Academic Excellence Scholarship, Ocean University of China	2018 - 2019

TEACHING EXPERIENCE

Teaching Assistant — Ice Sheets, Climate & Sea Level Rise, UTAS	Sep 2025 - Oct 2025
Designed and led practicals; assisted students in analysing model results	
Lecturer — Kioloa Winter School: Antarctic Tipping Points, ANU	Jun 2025
Delivered lectures on tipping points and irreversibility within ice sheets	

PROFESSIONAL SERVICE & DEVELOPMENT

Creator & Developer— 3D ICE	Feb 2026 - present
A web-based Interactive Cryosphere Explorer to visualise Antarctica and Greenland	
Peer Reviewer	Apr 2025
<i>Nature Communications</i>	
Co-convenor — WilkesMIP Project	Jan 2025 - present
Coordinating an international model intercomparison project	
Organiser and Host — UTAS Ice Sheets Group Meeting	Jun 2023 - Aug 2024
Organised weekly group seminars and facilitated discussions	
Karthus Summer School on Ice Sheets and Glaciers	May 2024
<i>Karthus, Italy</i>	
Monash Hackathon — Disentangling Uncertainties in ISMIP6-2300	Mar 2024
<i>Monash University, Australia</i>	
Visiting Research Student	Sep 2021 - Feb 2022
<i>Prof. John Moore's Group, Beijing Normal University, China</i>	

PRESENTATIONS

Oral Presentation , Antarctic Research Centre, Victoria University of Wellington	Feb 2026
Oral Presentation , Climate and Cryosphere Conference 2026, Wellington	Feb 2026
Oral Presentation , Antarctica Day Symposium, Hobart	Dec 2025
Oral Presentation , FRISP – Southern Hemisphere Workshop, Queensland	Jul 2025
Invited Seminar , School of Oceanography, Shanghai Jiao Tong University	Dec 2024
Poster , Australian Antarctic Research Conference, Hobart	Nov 2024
Oral Presentation , IMAS HDR Conference, Hobart	Nov 2024
Invited Seminar , ICEMAP Antarctic Modelling Workshop, Rovaniemi	May 2024
Poster , European Geosciences Union (EGU) General Assembly, Vienna	Apr 2024
Poster , AAPP Symposium, Hobart	Oct 2023
Poster , ACCESS Community Workshop, Canberra	Sep 2023
Oral Presentation , Coupled Ice-Sheet Modelling and Simulations Workshop	Feb 2023
Lightning talk , European Geosciences Union (EGU) General Assembly (Online)	Apr 2021

TECHNICAL SKILLS

Computing	MATLAB, Python, Fortran, Linux Shell, High-Performance Computing, Git/GitHub, HTML/CSS (Web Development)
Modelling	Elmer/Ice (Ice Flow), GlaDS (Subglacial Hydrology), ROMS (Ocean Circulation)
Software	QGIS, ParaView, Adobe Illustrator
Languages	English (Fluent), Mandarin Chinese (Native)